The Cone Generated by Positive Semidefinite Mean Polynomials

Mehdi Ghasemi, Salma Kuhlmann

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Abstract

Certain classes of homogeneous polynomials, such as Positive Semidefinite (PSD) Diagonal-minus-Tail (D-T) forms and those derived from the Arithmetic-Geometric Mean inequality—specifically Sums of Nonnegative Circuit (SONC) polynomials—can be identified as nonnegative mean polynomials. This paper studies PSD forms defined as weighted power means, denoted by $M_{q,p}(Y, w)$, and investigates the cone of nonnegative mean real n-variate forms of degree $2d$, denoted by $\mathcal{M}_{n,2d}(X, \alpha)$. We analyze the relationship between $\mathcal{M}_{n,2d}$ and other fundamental cones in real algebraic geometry: the cone of all PSD forms ($\mathcal{P}_{n,2d}$), the cone of Sums of Squares ($\Sigma_{n,2d}$), and the SONC cone ($\mathcal{C}_{n,2d}$). Furthermore, we examine the structural properties of the sum of the SOS and SONC cones, commonly referred to as the SOSONC cone.

1 Introduction

The certification of nonnegativity for multivariate polynomials is a central problem in real algebraic geometry ...with significant applications in polynomial optimization. While deciding whether a general polynomial is Positive Semidefinite (PSD) is NP-hard, specific sub-classes of polynomials offer tractable certificates of nonnegativity. Notable examples include Sums of Squares (SOS), Positive Semidefinite Diagonal-minus-Tail (PSD D-T) forms, and Sums of Nonnegative Circuit (SONC) polynomials, which are derived from the Arithmetic-Geometric Mean inequality.

In this paper, we propose a unified framework for these classes by identifying them with *nonnegative mean polynomials*. We study PSD forms that are constructed as weighted power means, denoted by $M_{q,p}(Y, w)$, where $Y = (Y_1, ..., Y_m)$ is a tuple of symbols and $w \in \mathbb{R}^m_+$ is a vector of positive weights.

Specifically, we investigate the **cone of nonnegative mean real n-variate forms of degree** $2d$, denoted as $\mathcal{M}_{n,2d}(X, \alpha)$, where $X = (X_1, ..., X_n)$ and $\alpha \in \mathbb{N}^n_0$. A primary objective of this work is to elucidate the relationship between $\mathcal{M}_{n,2d}$ and the following established cones:

- $\mathcal{P}_{n,2d}$: The cone of all positive semidefinite (PSD) forms. Note that by definition, $\mathcal{M}_{n,2d} \subseteq \mathcal{P}_{n,2d}$.
- $\Sigma_{n,2d}$: The cone of sums of squares (SOS) polynomials.
- $\mathcal{C}_{n,2d}$: The cone of sums of nonnegative circuit (SONC) polynomials.
- **SOSONC**: The sum of the SOS and SONC cones, defined by M. Schick as $(\Sigma + \mathcal{C})_{n,2d}$.

Structure of the Paper

The remainder of this paper is organized as follows:

In §2, we provide the preliminaries on mean polynomials. We introduce the weighted power means $M_{q,p}(Y, w)$ for $q, p \in \mathbb{N}_0$, and utilizing Jensen's inequality, we prove that if $p < q$, the form $M_{q,p}$ is PSD (Proposition 1). We further demonstrate how standard classes of nonnegative polynomials can be recovered as instances of these means.

In §3, we revisit nonnegative circuit polynomials. We provide necessary and sufficient conditions for the nonnegativity of D-T forms and circuit polynomials using the language of mean polynomials, recovering established results found in the literature. We formally define the SONC cone $\mathcal{C}_{n,2d}$ (Definition 1) and show that the circuit nonnegativity criterion (Theorem 2) implies the D-T criterion (Theorem 1) as a special case; the two results are not equivalent.

In §4, we investigate the relationship between mean polynomials and Sums of Squares. We study $M_{2d,p}(X, \alpha)$ with $|\alpha| = 2d$ and $p \mid 2d$. We prove that $M_{2d,0}$ forms are Sums of Binomial Squares (SOBS) (Proposition 2). Additionally, for cases $d = 1, 2, 3$, we prove via induction and reduction to Hilbert's theorem that these forms are SOS for all $n \in \mathbb{N}$ (Theorem 3).

In §5, we move beyond the SOSONC cone to understand the general form $M_{q,p}(Y, w)$. We examine the Choi-Lam form Q and the second Robinson form $\hat{R}$. While Q is known to be SONC but not SOS, we establish a representation of Q and $\hat{R}$ as mean polynomials—specifically showing that $\hat{R}$ can be expressed as $8M_{1,0}(G, \mathbf{1})$ for a specific linear transformation G. This identifies $\hat{R}$ as a separating form for the cone of mean polynomials and the SOSONC cone.

The topics of Positivestellensatz, polynomial optimization, and differential-algebra/KKT objectives (§6–§7) are reserved for a follow-up paper.

2 Preliminaries on Mean Polynomials

In this section, we introduce the concept of weighted power means and establishing the fundamental inequalities that allow us to construct nonnegative polynomials.

Let $Y = (Y_1, \ldots, Y_m)$ be an m-tuple of variables (or symbols) and $w = (w_1, \ldots, w_m) \in \mathbb{R}_+^m$ be a vector of positive weights. We denote the power mean of order p as $M_p(Y, w)$.

2.1 Jensen's Inequality and Monotonicity

The nonnegativity of the mean polynomials studied in this paper relies on the monotonicity of the power mean function. We prove this using Jensen's inequality.

Proposition 1. *If $p < q$, then $M_p(Y, w) \leq M_q(Y, w)$ for any non-negative input Y.*

Proof. Consider the function $\varphi(x) = x^{\frac{2q}{p}}$ with $q > p$. The second derivative of this function is:

$$\varphi''(x) = \frac{2q}{p}\left(\frac{2q}{p} - 1\right) x^{\frac{2q}{p}-2}$$

Since $q > p$, the coefficient $\frac{2q}{p}(\frac{2q}{p} - 1)$ is strictly positive. Consequently, $\varphi(x)$ is a convex function on $\mathbb{R}_+$.

By Jensen's inequality, for any convex function φ and weights w_i, we have:

$$\varphi\left(\frac{\sum w_i x_i}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i)}{\sum w_i}$$

Substituting x_i with x_i^p in the inequality above, we obtain:

$$\varphi\left(\frac{\sum w_i x_i^p}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i^p)}{\sum w_i}$$

Substituting the definition $\varphi(t) = t^{q/p}$ (noting that the factor of 2 in the degree is preserved in the structure of the resulting forms), this inequality simplifies to:

$$\left(\frac{\sum w_i x_i^p}{\sum w_i}\right)^{\frac{q}{p}} \leq \frac{\sum w_i x_i^q}{\sum w_i}$$

Taking the q-th root, this is equivalent to $M_p(Y, w) \leq M_q(Y, w)$. □

2.2 The Cone of Mean Polynomials

Using the inequality established above, we can define a class of nonnegative polynomials. By raising both sides of the inequality $M_p \leq M_q$ to the power $c = \mathrm{lcm}(q, p)$, we eliminate rational exponents. We define the **mean polynomial form** $\mathcal{M}_{q,p}$ as:

$$\mathcal{M}_{q,p}(Y, w) = M_q^c(Y, w) - M_p^c(Y, w)$$

By Proposition 1, $\mathcal{M}_{q,p}$ is a nonnegative polynomial function in variables Y for all $q > p$.

Properties of Power Means

We recall the following basic limiting properties of the mean function $M_p(x, w)$ that are relevant to the boundaries of these cones:

- **Minimum:** $\lim_{p\to-\infty} M_p(x,w) = \min(x_1,\dots,x_n)$.
- **Maximum:** $\lim_{p\to\infty} M_p(x,w) = \max(x_1,\dots,x_n)$.
- **Geometric Mean:** $\lim_{p\to 0} M_p(x,w) = (\prod_{i=1}^n x_i^{w_i})^{1/\sum w_i}$.

Finally, we note that the equality $M_p(x,w) = M_q(x,w)$ holds if and only if all elements are equal, i.e., $x_1 = x_2 = \cdots = x_n$.

3 Nonnegative Diagonal minus Tail and Circuit Polynomials

In this section, we investigate the conditions for nonnegativity of Diagonal-minus-Tail (D-T) forms and general Circuit polynomials. We utilize the language of mean polynomials to re-derive established results found in [5] and [1].

Theorem 1. *Let $f(x)$ be a Diagonal-minus-Tail form defined as:*

$$f(x) = \sum_{i=1}^n \beta_i x_i^{2d} - \mu x^\alpha$$

where:

- $x = (x_1,\dots,x_n) \in \mathbb{R}^n$*;*
- *coefficients satisfy* $\beta_i, \mu > 0$*;*
- *exponents satisfy* $\alpha = (\alpha_1,\dots,\alpha_n) \in \mathbb{N}_0^n$ *and*

$$\sum_{i=1}^n \alpha_i = 2d.$$

The function $f(x)$ is positive semidefinite (PSD) on $\mathbb{R}_+^n$ if and only if:

$$\mu^{2d} \prod_{i=1}^n \alpha_i^{\alpha_i} \le (2d)^{2d} \prod_{i=1}^n \beta_i^{\alpha_i}$$

Proof. The proof relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality.
($\Leftarrow$) Assume the condition holds. To show $f(x) \ge 0$, it suffices to show $\sum_{i=1}^n \beta_i x_i^{2d} \ge \mu x^\alpha$. Apply the weighted AM-GM inequality to the terms $y_i = \frac{\beta_i}{\alpha_i} x_i^{2d}$ with weights $w_i = \alpha_i$ (noting $\sum w_i = 2d$). We obtain:

$$\frac{\sum_{i=1}^n \alpha_i \left(\frac{\beta_i}{\alpha_i} x_i^{2d}\right)}{\sum_{i=1}^n \alpha_i} \ge \left(\prod_{i=1}^n \left(\frac{\beta_i}{\alpha_i} x_i^{2d}\right)^{\alpha_i}\right)^{1/2d}$$

Simplifying the LHS gives $\frac{1}{2d}\sum \beta_i x_i^{2d}$. Rearranging yields:

$$\sum_{i=1}^{n} \beta_i x_i^{2d} \geq 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d} x^{\alpha}$$

By the hypothesis, $\mu \leq 2d \prod (\frac{\beta_i}{\alpha_i})^{\frac{\alpha_i}{2d}}$. Thus, $\sum \beta_i x_i^{2d} \geq \mu x^{\alpha}$.
(⇒) Conversely, assume $f(x)$ is PSD. We prove the condition by contradiction. Suppose:

$$\mu > 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d}$$

We construct a test point $x^* \in \mathbb{R}_+^n$ where the AM-GM equality holds. Let $x_i^* = (\frac{\alpha_i}{\beta_i})^{1/2d}$. Evaluating the sum term:

$$\sum_{i=1}^{n} \beta_i (x_i^*)^{2d} = \sum_{i=1}^{n} \beta_i \left(\frac{\alpha_i}{\beta_i} \right) = \sum_{i=1}^{n} \alpha_i = 2d$$

Evaluating the monomial term:

$$(x^*)^{\alpha} = \prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\frac{\alpha_i}{2d}} = \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

Substituting into $f(x^*)$:

$$f(x^*) = 2d - \mu \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

Under the assumption that the condition fails (strictly greater μ), we have $f(x^*) < 0$, contradicting that f is PSD. □

We now generalize this result to circuit polynomials, where the tail term is supported by a convex combination of the exponents of the diagonal terms.

Theorem 2. *Let $\alpha(0), \ldots, \alpha(r) \in \mathbb{N}^n$ be exponent vectors. Let $\beta \in \mathbb{N}^n$ be a convex combination of the $\alpha(j)$ vectors, such that $\beta = \sum_{j=0}^{r} \lambda_j \alpha(j)$ with $\lambda_j > 0$ and $\sum \lambda_j = 1$. Consider the polynomial $f(X)$ with coefficients $f_{\alpha(j)} > 0$:*

$$f(X) = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} - f_{\beta} X^{\beta}$$

The polynomial $f(X)$ is PSD for all $X \in \mathbb{R}_+^n$ if and only if:

$$|f_{\beta}| \leq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j}$$

Proof. Let $Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)}$ for $j = 0, \ldots, r$. By the weighted AM-GM inequality:

$$\sum_{j=0}^{r} \lambda_j Y_j \geq \prod_{j=0}^{r} Y_j^{\lambda_j}$$

Substituting Y_j:

$$\sum_{j=0}^{r} \lambda_j \left(\frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} \right) \geq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} \right)^{\lambda_j}$$

The LHS simplifies to $\sum f_{\alpha(j)} X^{\alpha(j)}$. The RHS simplifies to:

$$\left(\prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} \right) X^{\sum \lambda_j \alpha(j)} = \left(\prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} \right) X^{\beta}$$

Thus, if $|f_\beta| \leq \prod (\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}$, it follows that $\sum f_{\alpha(j)} X^{\alpha(j)} \geq |f_\beta| X^\beta \geq f_\beta X^\beta$, proving nonnegativity. The necessary condition follows by selecting X such that the variance of terms Y_j vanishes. □

Definition 1. *A polynomial whose coefficients satisfy the condition in Theorem 2 is called a* ***nonnegative circuit polynomial****. The* ***SONC cone*** $\mathcal{C}_{n,2d}$ *is the conic hull of all nonnegative circuit polynomials of degree* $2d$ *in* n *variables.*

Remark 1. *From the proof of the theorem 2, it is evident that every nonnegative circuit polynomial can be expressed as a mean polynomial of the form* $M_{1,0}(Y, w)$*, where* $Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)}$ *and* $w_j = \lambda_j$*.*

Definition 2. *The* ***SOSONC cone*** $(\Sigma + \mathcal{C})_{n,2d}$ *is the Minkowski sum of the SOS cone* $\Sigma_{n,2d}$ *and the SONC cone* $\mathcal{C}_{n,2d}$*, as introduced in [3].*

3.1 Diagonal-minus-Tail vs Circuit Forms

Theorem 2 implies Theorem 1 as a special case; the converse does not hold in general, so the two results are *not* equivalent. Every D-T form is a circuit polynomial (with a single tail term), but circuit polynomials allow an arbitrary number of diagonal terms whose exponents form a simplex, which is a strictly more general setting.

To see the implication, map the D-T form $f(x) = \sum \beta_i x_i^{2d} - \mu x^\alpha$ to the general framework:

- The diagonal terms correspond to $\alpha(i) = 2de_i$ with coefficients $f_{\alpha(i)} = \beta_i$.
- The tail term exponent is $\beta = \alpha$.
- The convex combination weights are $\lambda_i = \frac{\alpha_i}{2d}$. Hence

$$\sum \lambda_i = \frac{1}{2d} \sum \alpha_i = 1.$$

Applying the condition from Theorem 2:

$$\mu \leq \prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i/2d} \right)^{\alpha_i/2d} = \left(\prod_{i=1}^{n} \left(\frac{2d\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{\frac{1}{2d}}$$

Raising both sides to the power $2d$ yields $\mu^{2d} \leq (2d)^{2d} \prod (\frac{\beta_i}{\alpha_i})^{\alpha_i}$, recovering the condition in Theorem 1.

4 Investigation of $M_{2d,p}(X, \alpha)$ and Connection to SOBS and SOS

In this section, we investigate the containment of the mean polynomial forms $M_{2d,p}(X, \alpha)$ within the cones of Sums of Binomial Squares (SOBS) and Sums of Squares (SOS).

We first address the case where $p = 0$, which corresponds to the geometric mean.

Proposition 2. *The form $M_{2d,0}(X, \alpha)$ is a PSD Diagonal-minus-Tail (D-T) form and is a Sum of Binomial Squares (SOBS) for any $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 2d$.*

Proof. The fact that $M_{2d,0}$ is a PSD D-T form follows from the definitions established in Section 2. The proof that such forms are SOBS is established in [5]. □

We now consider the cases where $p > 0$. We prove that for small degrees $2d \in \{2, 4, 6\}$, these forms are not only PSD but also SOS.

Theorem 3. *The forms $M_{2d,p}(X, \alpha)$ are Sums of Squares (SOS) for all $n \in \mathbb{N}$ and any $\alpha \in \mathbb{N}^n$ with $|\alpha| = 2d$, for the cases:*

- $d = 1$ *(Quadratic forms), $p \in \{0, 1\}$;*
- $d = 2$ *(Quartic forms), $p \in \{0, 1, 2\}$;*
- $d = 3$ *(Sextic forms), $p \in \{0, 1, 2, 3\}$.*

Proof. We analyze each degree $2d$ separately.
Case $d = 1$ (Degree 2): The forms $M_{2,0}$ and $M_{2,1}$ are PSD quadratic forms. By Hilbert's 1888 Theorem, every nonnegative quadratic form is a sum of squares. Thus, they are SOS.
Case $d = 2$ (Degree 4): For $p = 0$, the form is SOBS (and thus SOS) by Proposition 2. We analyze $p = 1$ and $p = 2$.

- **$p = 1$:** The form is $M_{4,1}(X, \alpha) = \frac{1}{4} \sum \alpha_i X_i^4 - (\frac{1}{4} \sum \alpha_i X_i)^4$.
    1. For $n = 2$ (binary quartics) and $n = 3$ (ternary quartics), Hilbert's Theorem states that all PSD forms are SOS. Since $M_{4,1}$ is PSD, it is SOS.

2. For $n = 4$, we consider the uniform weights $\alpha = (1, 1, 1, 1)$. SDP decomposition using Irene [8] confirms $M_{4,1}(X, (1, 1, 1, 1))_{\text{sos}} = 0$. Moreover, an approximate Gram matrix for this instance is given in [8]:

$$G = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.25 & -0.01 & -0.04 & -0.01 & -0.04 & -0.04 & -0.01 & -0.04 & -0.04 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.05 & -0.01 & 0.01 & 0.01 & -0.04 & 0.01 & 0.01 & -0.02 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.01 & 0.25 & -0.04 & -0.01 & -0.04 & -0.04 & -0.01 & -0.04 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.01 & -0.04 & 0.05 & 0.01 & -0.01 & 0.01 & -0.02 & 0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & 0.01 & -0.01 & 0.01 & 0.05 & -0.01 & -0.02 & 0.01 & 0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.04 & -0.04 & -0.01 & -0.01 & 0.25 & -0.04 & -0.04 & -0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.01 & -0.04 & 0.01 & -0.02 & -0.04 & 0.05 & 0.01 & 0.01 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & 0.01 & -0.01 & -0.02 & 0.01 & -0.04 & 0.01 & 0.05 & 0.01 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.02 & -0.04 & 0.01 & 0.01 & -0.01 & 0.01 & 0.01 & 0.05 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.04 & -0.04 & -0.04 & -0.04 & -0.04 & -0.01 & -0.01 & -0.01 & 0.25
\end{bmatrix}$$

For non-uniform weights with $|\alpha| = 4$ (e.g., $(2, 1, 1, 0)$), the form effectively involves fewer than 4 variables, reducing to the $n \leq 3$ case, which is SOS.

3. For $n \geq 5$, any partitions of 4 into n integers must contain zeros. The form reduces to a case with $n \leq 4$, which we have shown to be SOS.

- $p = 2$**:** The form is $M_{4,2}(X, \alpha) = \frac{1}{4} \sum \alpha_i X_i^4 - (\frac{1}{4} \sum \alpha_i X_i^2)^2$. Similar to the $p = 1$ case:

1. For $n \leq 3$, it is SOS by Hilbert's Theorem.
2. For $n = 4$ with $\alpha = (1, 1, 1, 1)$, SDP decomposition using Irene [8] confirms $f_{\text{sos}} = 0$. The following is an approximate Gram matrix [8]:

$$G = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.19 & -0 & -0.06 & 0 & -0 & -0.06 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & -0 & 0 & -0 & 0 & -0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & -0 & 0.19 & 0 & -0 & -0.06 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & -0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0 & -0 & -0 & -0 & 0 & 0 & 0 & 0 & 0 & -0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & 0 & -0.06 & -0 & 0 & 0.19 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & -0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & 0 & -0.06 & 0 & -0 & -0.06 & 0 & 0 & 0 & 0.19
\end{bmatrix}$$

3. For $n \geq 5$, the support of α limits the effective variables to at most 4, reducing to the previous cases.

Case $d = 3$ **(Degree 6):** We examine $M_{6,p}(X, \alpha)$ for $p \in \{0, \ldots, 3\}$.

1. $n = 2$ **(Binary Sextics):** By Hilbert's Theorem, all PSD binary forms of degree 6 are SOS.

2. $n = 3$ **(Ternary Sextics):**
   - For $p = 0$, it is SOBS (Prop 2).
   - For $p = 1$, we analyze partitions of 6 into 3 parts: $(2, 2, 2)$, $(3, 2, 1)$, $(4, 1, 1)$, $(5, 1, 0)$, $(6, 0, 0)$. The case $\alpha = (2, 2, 2)$ is confirmed to be SOS via SDP decomposition using Irene [8]. Cases with a zero component reduce to $n \leq 2$ (binary sextics, covered by Hilbert's Theorem). The remaining cases $(3, 2, 1)$ and $(4, 1, 1)$ are also confirmed to be SOS via SDP decomposition using Irene [8].
   - For $p = 2$ and $p = 3$, SOS membership for all partitions of 6 into 3 non-zero parts is confirmed via SDP decomposition using Irene [8]; cases with zero components reduce to $n \leq 2$.
3. $n = 4$**:** Relevant partitions of 6 include $(2, 2, 1, 1)$ and $(3, 1, 1, 1)$. SDP decomposition using Irene [8] confirms these specific instances are SOS. Partitions containing zeros reduce to $n \leq 3$.
4. $n \geq 5$**:** Any partition of 6 into $n \geq 6$ parts (e.g., $(1, 1, 1, 1, 1, 1)$) or parts with zeros reduces the effective number of variables.

This completes the proof for $d = 1, 2, 3$. □

Remark 2. *Two open questions remain regarding these forms:*

1. *Does Theorem 3 generalize to any degree d?*
2. *Are these forms contained within the SONC cone for all parameters?*

A clean degree-8 pilot computed with Irene [8] for $d = 4$, $n \in \{3, 4\}$, $p \in \{0, 1, 2, 4\}$, and representative nondegenerate template families yields a consistent pattern: every tested nondegenerate instance admitted an SDP-based SOS decomposition, while SONC feasibility held only for the $p \in \{0, 4\}$ slices. For the same families with $p \in \{1, 2\}$, the SONC signomial constraints remained infeasible under local and global solves and under error bounds 10^{-6}, 10^{-8}, and 10^{-10}. This provides positive evidence for a degree-8 extension of Theorem 3 and negative evidence against a broad SONC containment statement, while still leaving a theorem-level explanation open.

Scoped extension conjecture for Theorem 3. *The current computational record supports the following manuscript-ready scoped extension of Theorem 3. For nondegenerate tested families of $M_{2d,p}(X, \alpha)$, SOS feasibility is computationally confirmed on two slices: first, the frozen $d \in \{4, 5\}$, $n \in \{3, 4\}$ grid across the executed p-values and tolerance settings; second, the tested partial $d = 6$, $n = 3$ uniform L-C1 slice. These confirmed regions are recorded in* `phase3_pilot_summary_clean.md`, `phase3_pilot_summary_clean_d5.md`, `phase3_runs_clean_d6_lc1_batch1.jsonl`, `phase3_runs_clean_d6_lc1_batch2.jsonl`, *and* `phase3_pilot_summary_clean_d6_lc1_batch2.md`.

We therefore state B1 in the following scoped form: the SOS behavior of Theorem 3 extends beyond $d \leq 3$ on the confirmed slices above, and it is reasonable to conjecture that this extension persists on a broader structured class of mean-polynomial families. However, that broader claim is not promoted to theorem-level status for the deferred/open slice $d = 6$, $n \geq 4$. On that slice, the current SDP stack yields only solver-ceiling behavior: the $n = 4$ probes (uniform, boundary, mixed templates) time out under budgets up to 1800s, and the $n = 5$ escalation sequence `CVXOPT` $\rightarrow$ `SDPA` $\rightarrow$ `CSDP` *produces 18/18 timeouts at both 600s and 1200s per solver with no SOS certificate and no hard infeasibility certificate. The deferred evidence is archived in* `phase3_pilot_summary_clean_d6_lc1_batch3.md`, `phase3_pilot_summary_clean_d6_lc1_batch4_boundary.md`, `phase3_pilot_summary_clean_d6_lc1_batch5_mixed.md`, `phase3_pilot_summary_clean_d6_lc1_batch6_uniform_p6.md`, `phase3_pilot_summary_clean_d6_lc1_batch7_boundary_p6.md`, `phase3_pilot_summary_clean_d6_lc1_batch8_mixed_p6.md`, *and* `phase3_pilot_summary_d6_lc1_escalation_pass2.md`.*

Accordingly, B1 should be read as a scoped conjectural extension with explicit boundaries: computationally confirmed on the tested nondegenerate $d \in \{4, 5\}$ slices and the partial $d = 6$, $n = 3$ slice, but explicitly deferred/open for $d = 6$, $n \geq 4$ under the present solver ceiling. In the current project cycle, we therefore treat this slice as computationally frozen rather than mathematically resolved: the existing evidence supports backend-limited deferment, not a negative theorem. Reopening this thread requires either access to a materially stronger SDP backend or a structure-aware reformulation that changes the computational profile without changing the mathematical claim.

Scoped SONC conjecture. *The current computational record does not support a broad degree-only SONC containment statement for the families $M_{2d,p}(X, \alpha)$. Instead, the evidence favors a support-sensitive formulation in terms of an exponent-interiority margin. Define*

$$\Delta_{\text{support}}(d, p) := \frac{\min\{p, d - p\}}{d}.$$

Empirically, on the executed nondegenerate slices, SONC status aligns with the sign of Δ_{support}: boundary cases ($\Delta_{\text{support}} = 0$) are SONC-feasible, while interior cases ($\Delta_{\text{support}} > 0$) are robustly SONC-infeasible under the executed diagnostic reruns.

Conjecture 1 (Delta-support criterion on tested mean-polynomial slices)**.** *For the tested nondegenerate families $M_{2d,p}(X, \alpha)$ in the executed slices below, SONC status is support-sensitive and consistent with $\Delta_{\text{support}}(d, p)$:*

1. *clean pilot $d \in \{4, 5\}$, $n \in \{3, 4\}$,*
2. *structural-stress diagnostics at $d = 6$,*
3. *nondegenerate SONC rows from the executed $d = 6$ L-C2 clean pilot.*

On these slices, $\Delta_{\text{support}} = 0$ aligns with observed SONC feasibility, and $\Delta_{\text{support}} > 0$ aligns with robust SONC-infeasibility under the executed local/global and tightened-tolerance diagnostics.

Scope caveat. Conjecture 1 is an empirical, slice-local statement tied to the executed diagnostics; it is not yet a universal criterion for all $M_{2d,p}(X, \alpha)$ families. The remaining theorem-stage task is an analytic derivation (or sharp delimitation) of this criterion via circuit/barycentric arguments.

Computational Evidence Snapshot (Degrees 8 and 10)

We summarize the clean Irene-based pilot runs for $d \in \{4, 5\}$, $n \in \{3, 4\}$, $p \in \{0, 1, 2, 4, 5\}$ with template families generated from balanced, mixed, and boundary weight patterns (one-hot degeneracies tracked separately as inconclusive). The clean records are reported in:

- `phase3_runs_clean.jsonl`
- `phase3_runs_clean_d5.jsonl`
- `phase3_classification_table.csv`

For both $d = 4$ and $d = 5$, every nondegenerate tested instance admitted an SDP-based SOS decomposition in our current grid. In contrast, SONC feasibility is support-sensitive: the nondegenerate families with $p \in \{1, 2\}$ that fail at degree 8 remain infeasible under local/global SONC solves and tighter tolerance settings, while GP relaxations remain feasible on all tested nondegenerate families in the same grid. This pattern supports a refined view in which SOS positivity extends robustly on the tested families, whereas SONC membership requires stronger structural conditions than the broad conjectural form. Tables 1 and 2 summarize the refreshed classification output using the corrected support labels. We retain a placeholder row for $d \geq 6$ in the manuscript tables; a completed $d = 6$ L-C2 SONC/GP slice and a staged $d = 6$ L-C1 SOS/SDP extension are tracked in companion artifacts. For $d = 6$, $n = 3$ uniform L-C1 slices are successful in the tested grid, whereas $n = 4$ L-C1 slices exhibit timeout-dominated behavior across uniform, boundary, and mixed templates (including $p = 6$ cross-template probes) under budgets up to 1800s. We therefore mark $d = 6$ L-C1 as computationally deferred in the current manuscript tables. Cross-degree SONC diagnostics now show a consistent zero-recovery pattern on every executed clean-pilot failure set: all 24 nondegenerate SONC failures at $d = 4$ remain failures under local/global and tightened-tolerance reruns, the same 24/24 pattern persists at $d = 5$, and the refreshed $d = 6$ L-C2 clean slice expands this to 48/48 robust failures across the uniform, boundary, and mixed templates for $n \in \{3, 4\}$. This strengthens the interpretation that the observed SONC negatives are support-structural in the tested slices rather than artifacts of the local/global solve choice or loose error tolerances. Additional CX-1 tractability probes indicate a solver-scaling ceiling for the current CVXOPT SDP pipeline at $n \leq 4$ for L-C1 families: d=4 probes at $n = 5, 6$

(boundary/mixed templates) timeout at 120s, d=5 at $n = 5$ (boundary probe) times out at 900s, and even d=3 at $n = 5$ requires 272s for a single successful boundary probe. This supports treating $n \geq 5$ high-degree sparse-family extensions as computationally deferred pending specialized solver paths. An initial CX-2 transform-stress micro-batch on Choi-Lam/Robinson-derived families shows transform-sensitive SONC behavior: Choi-Lam base/perturbed slices are feasible for SONC and GP, Robinson-hat base is infeasible for both methods, and the transformed Robinson-hat base becomes SONC-feasible while remaining GP-infeasible. This provides first concrete evidence that linear transform choices can change SONC classification behavior in the tested slice. A follow-up CX-2 coefficient sweep sharpens this picture: for Choi-Lam tail coefficients $c \in \{4.0, 3.95, 3.9, 3.8\}$, SONC and GP remain feasible throughout, while for Robinson-hat coefficients $c \in \{2.0, 1.98, 1.95, 1.9, 1.85, 1.8\}$ the pre-transform forms remain SONC/GP-infeasible. Post-transform SONC feasibility appears only at the baseline Robinson-hat coefficient $c = 2.0$, with nearby coefficients in this sweep remaining SONC-infeasible. In the current computational window, this supports a localized transform-sensitive boundary effect rather than a broad coefficient-robust transform classification shift.

Localized transform-boundary conjecture (B3, CX-2). The current CX-2 record motivates a narrowly scoped transform-boundary conjecture: within the tested Robinson-hat family and current normalization, the observed post-transform SONC-feasible point at baseline coefficient $c = 2.0$ behaves as a localized boundary phenomenon rather than as evidence of a coefficient-robust transform-induced SONC phase shift. Equivalently, the present data support the existence of a small transform-sensitive region around the baseline where SONC classification can flip, while nearby coefficients in the tested sweep remain in the SONC-failure regime. We therefore treat B3 as a localized conjectural statement tied to this family and slice, not as a global transform principle.

Two targeted OP3 sweeps further sharpen this picture. A local 7×7 crossed sweep over coefficient perturbations

$$\begin{aligned} c &\in \{1.9925, 1.9950, 1.9975, 2.0000, 2.0025, 2.0050, 2.0075\}, \\ t &\in \{0.90, 0.94, 0.97, 1.00, 1.03, 1.06, 1.10\}. \end{aligned}$$

This 49-point grid finds that only the exact center point $(c, t) = (2.0, 1.0)$ is SONC-robust-success; every non-center perturbation is robust-fail across all four SONC diagnostic configurations (gates: H1 Fail, H2 Pass, H3 Pass). A subsequent wider coefficient scan at fixed $t = 1.0$ over

$$\begin{aligned} c \in \{&1.5, 1.6, 1.7, 1.8, 1.85, 1.9, 1.95, 1.98, 1.99, 1.995, \\ &2.0, 2.005, 2.01, 2.02, 2.05, 2.1, 2.2, 2.5\} \end{aligned}$$

confirms that $c = 2.0$ is the unique SONC-success point in the tested range: all 17 other coefficient values fail robustly across all diagnostic reruns. This point-level sharpness is consistent with the Robinson-hat form being exactly at

Table 1: Pilot classification summary for $d = 4$ (degree 8). Entries report (SDP/SONC/GP) using S = success, T = timeout, F = fail, and I = inconclusive.

template	representative α	support class	n values	$p = 0$	$p = 1$	$p = 2$	$p = d$
uniform	$n = 3$: $(3, 3, 2)$; $n = 4$: $(2, 2, 2, 2)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
mixed	$n = 3$: $(4, 2, 2)$; $n = 4$: $(4, 2, 2, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
boundary	$n = 3$: $(7, 1, 0)$; $n = 4$: $(7, 1, 0, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
sparse	$n = 3$: $(8, 0, 0)$; $n = 4$: $(8, 0, 0, 0)$	degenerate	$\{3, 4\}$	I/I/I	I/I/I	I/I/I	I/I/I
$d \geq 6$	–	placeholder	$\{3, 4\}$ planned	deferred	deferred	deferred	deferred

Table 2: Pilot classification summary for $d = 5$ (degree 10). Entries report (SDP/SONC/GP) using S = success, T = timeout, F = fail, and I = inconclusive.

template	representative α	support class	n values	$p = 0$	$p = 1$	$p = 2$	$p = d$
uniform	$n = 3$: $(4, 3, 3)$; $n = 4$: $(3, 3, 2, 2)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
mixed	$n = 3$: $(5, 3, 2)$; $n = 4$: $(5, 3, 2, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
boundary	$n = 3$: $(9, 1, 0)$; $n = 4$: $(9, 1, 0, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
sparse	$n = 3$: $(10, 0, 0)$; $n = 4$: $(10, 0, 0, 0)$	degenerate	$\{3, 4\}$	I/I/I	I/I/I	I/I/I	I/I/I
$d \geq 6$	–	placeholder	$\{3, 4\}$ planned	deferred	deferred	deferred	deferred

the boundary of the SONC certificate cone under the tested transform. Promotion beyond conjectural status requires mathematical characterisation of this boundary (e.g. via Newton-polytope or AM/GM structure analysis) rather than further solver perturbation studies. For deferred $d = 6/n \geq 5$ L-C1 SOS slices, we prepared and executed a specialized-solver escalation route in the benchmark driver via solver-sequence fallback (`cvxopt` $\rightarrow$ `sdpa` $\rightarrow$ `csdp`) with per-solver timeout caps and per-attempt logging. Pass1 (600s per solver) on the 18 nondegenerate $n = 5$ cases (uniform, boundary, mixed templates) returned 18/18 timeouts across all three solvers; pass2 (1200s per solver) on the same 18 cases again returned 18/18 timeouts with no success and no hard-fail certificates. Thus, doubling the cap produced no solver/template delta, and under the current interior-point SDP stack this $d = 6$, $n \geq 5$ L-C1 slice remains computationally intractable within practical budgets. We therefore retain this slice as deferred in the manuscript tables and treat it as a solver-ceiling computational limitation rather than a theorem-level negative result.

Deferred higher-degree slice. The $d \geq 6$ rows are retained as placeholders for a future extension of the same protocol. No higher-degree computations are included in the present manuscript snapshot.

Claim-Status Synthesis (Frozen $d = 4, d = 5$ Slice)

Using the frozen clean artifacts and refreshed classification outputs (listed in the reproducibility note below), we classify the current computational evidence as follows.

- **Confirmed (within tested grid).** For nondegenerate tested families at $d = 4, 5$ with $n \in \{3, 4\}$, SDP/SOS and GP outcomes are consistently feasible in the current runs.

- **Weakened (scope-limited).** Any broad SONC containment formulation for the same families is not supported uniformly; the evidence favors a support-sensitive SONC statement rather than a degree-only claim.

- **Negative (robust failures).** For nondegenerate families with $p \in \{1, 2\}$, SONC infeasibility persists under local/global solves and tighter tolerances in both degree slices, while SDP/SOS and GP remain feasible.

This synthesis is tied to the classification tables in Tables 1 and 2, with a partial higher-degree extension from the completed $d = 6$ L-C2 SONC/GP slice and a deferred-status $d = 6$ L-C1 SOS/SDP record.

Confirmed hypotheses (artifact-scoped). Within the tested computational window, the following claims are supported by direct evidence. Each claim is intentionally scoped by degree/variable/template slice.

- **C1 (SOS robustness on the frozen slice).** For nondegenerate template families at $d \in \{4, 5\}$, $n \in \{3, 4\}$, and $p \in \{0, 1, 2, 4, 5\}$, SDP/SOS decomposition is consistently feasible in the current runs. Evidence: frozen clean summaries for $d = 4$ and $d = 5$, plus the consolidated classification table artifact.

- **C2 (GP feasibility on the same frozen slice).** On the same nondegenerate $d \in \{4, 5\}$ tested grid, GP relaxations remain feasible across all recorded cases in the refreshed classification outputs. Evidence: the same frozen clean summaries and the refreshed classification table.

- **C3 (partial higher-degree SOS extension at low-n).** For the tested $d = 6$ L-C1 uniform $n = 3$ slice, SOS/SDP outcomes are feasible across the executed p grid. Evidence: d=6 L-C1 batch1/batch2 JSONL artifacts and the corresponding clean-summary note for batch2.

Weakened or null findings (scope-limited). The following outcomes narrow the current conjectural scope or report null gains under stronger computational budgets.

- **W1 (broad SONC containment is weakened).** For nondegenerate tested families at $d \in \{4, 5\}$ with $n \in \{3, 4\}$, SONC infeasibility persists on the $p \in \{1, 2\}$ slices under local/global solves and tighter tolerances. Implication: SONC claims should be formulated as support-sensitive, not degree-only. Evidence: SONC diagnostics summaries for $d = 4$ and $d = 5$. Caveat: this is not a theorem-level non-membership proof for all families.

- **W2 (null improvement from escalation pass1 to pass2).** For $d = 6$, $n = 5$, L-C1 nondegenerate templates (uniform/boundary/mixed), increasing the per-solver cap from 600s to 1200s in the `cvxopt` $\rightarrow$ `sdpa` $\rightarrow$ `csdp` sequence yields no first success and no hard-fail certificate. Implication: current conclusions are solver-ceiling computational limitations, not

mathematical refutations of SOS membership. Evidence: escalation pass1 and pass2 summary artifacts. Caveat: this does not prove infeasibility in stronger solver stacks.

- **W3 (structural SONC diagnostics remove tuning-artifact explanation).** Across the executed clean-pilot ladder, alternate SONC diagnostic configurations (local/global and tighter tolerance) show zero recoveries on every failed nondegenerate case: 24/24 at $d = 4$, 24/24 at $d = 5$, and 48/48 on the refreshed $d = 6$ L-C2 slice. Implication: these failures should be treated as formulation-robust negatives in the tested slices. Evidence: SONC diagnostics summaries for $d = 4$, $d = 5$, and the refreshed $d = 6$ L-C2 slice. Caveat: robustness is empirical on the executed slices, not a universal impossibility theorem.

Counterexamples and anomalies register (follow-up tracked). To keep negative evidence first-class, we maintain the following compact register with fields {family/slice, status pattern, suspected cause, confidence, follow-up action}.

- **A1 (CX-2 transform-sensitive boundary behavior).** Family/slice: Robinson-hat transform probe and coefficient sweep. Status pattern: pre-transform infeasible, baseline post-transform SONC-feasible, nearby transformed coefficients reverting to SONC-infeasible. Suspected cause: localized geometric sensitivity of the transformed support/coefficient regime, not a uniform transform effect. Confidence: medium (replicated in two batches on a narrow coefficient band). Follow-up action: run a denser local coefficient sweep around the baseline and test adjacent transform families. Evidence: CX-2 transform-probe summary artifacts from batch1 and batch2.

- **A2 ($d = 6$, $n = 5$ L-C1 solver-ceiling timeout wall).** Family/slice: nondegenerate uniform, boundary, and mixed templates with the index set
$$p \in \{0, 1, 2, 3, 4, 6\}.$$
Status pattern: all-timeout in pass1 (600s/solver) and pass2 (1200s/solver) across the CVXOPT, SDPA, and CSDP sequence. Suspected cause: SDP size/conditioning driven scalability ceiling under the current interior-point stack. Confidence: high (18/18 consistent timeouts in both passes, no countervailing solver signal). Follow-up action: keep this slice deferred in theorem claims and revisit only with stronger solvers or structure-exploiting formulations. Evidence: pass2 JSONL artifacts, the escalation pass2 summary, and the theorem ledger record.

Reproducibility note. The tables above are derived from the frozen $d = 4, d = 5$ artifacts:

- `phase3_runs_clean.jsonl`

- `phase3_runs_clean_d5.jsonl`
- `phase3_classification_table.csv`
- `phase3_classification_table.md`

A partial higher-degree extension is available for $d = 6$. The completed L-C2 SONC/GP slice and staged L-C1 SOS/SDP artifacts are:

- `phase3_runs_clean_d6_lc2.jsonl`
- `phase3_pilot_summary_clean_d6_lc2.md`
- `phase3_sonc_diagnostics_d6_lc2.jsonl`
- `phase3_sonc_diagnostics_d6_lc2.md`
- `phase3_runs_clean_d6_lc1_batch1.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch1.md`
- `phase3_runs_clean_d6_lc1_batch2.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch2.md`
- `phase3_runs_clean_d6_lc1_batch3.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch3.md`
- `phase3_runs_clean_d6_lc1_batch4_boundary.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch4_boundary.md`
- `phase3_runs_clean_d6_lc1_batch5_mixed.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch5_mixed.md`
- `phase3_runs_clean_d6_lc1_batch6_uniform_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch6_uniform_p6.md`
- `phase3_runs_clean_d6_lc1_batch7_boundary_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch7_boundary_p6.md`
- `phase3_runs_clean_d6_lc1_batch8_mixed_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch8_mixed_p6.md`
- `phase3_runs_clean_cx1_d4_n5.jsonl`
- `phase3_runs_clean_cx1_d4_n6.jsonl`
- `phase3_runs_clean_cx1_batch1_d5_n5_boundary_mixed.jsonl`

- `phase3_runs_clean_cx1_d3_n5_probe.jsonl`
- `phase3_pilot_summary_cx1_d3_n5_probe.md`
- `phase3_runs_clean_cx2_transform_probe.jsonl`
- `phase3_pilot_summary_cx2_transform_probe.md`
- `phase3_runs_clean_cx2_transform_probe_batch2.jsonl`
- `phase3_pilot_summary_cx2_transform_probe_batch2.md`
- `phase3_runs_clean_d6_lc1_escalation_smoke.jsonl`

The manuscript snippet `results/phase3_classification_tables.tex` is generated from the refreshed CSV by the table-builder script in `scripts/`. This keeps updates to the frozen $d = 4, d = 5$ slice and staged $d \geq 6$ extensions reproducible without manual table editing.

Follow-up evidence. The CX-1 tractability probe is completed and currently supports a solver-scaling bottleneck interpretation (rather than a new non-SOS counterexample) for sparse-family $n \geq 5$ slices under the present CVXOPT SDP setup. A first CX-2 transform-stress batch is also completed, showing a Robinson-hat base pre/post transform SONC-status change in the tested setup; a second CX-2 coefficient sweep indicates this behavior is currently localized near the baseline Robinson-hat coefficient under the tested transform family. Specialized-solver escalation paths for deferred $d = 6/n \geq 5$ SOS/SDP slices are now implemented and ready for full pass execution.

Phase 4 Integration Kickoff: Open Problems and Conjecture Backlog

To initiate Phase 4 integration, we record a ranked open-problem list and a conjecture backlog that directly links theory-facing statements to the completed computational evidence. In addition, we now rank candidate publishable claims from strongest to most exploratory.

1. **Publishable Claim PC1 (strongest):** on the tested nondegenerate $d \in \{4, 5\}$ slice with $n \in \{3, 4\}$, SOS/SDP and GP are consistently feasible while SONC is support-sensitive with robust failures on the $p \in \{1, 2\}$ families under diagnostic perturbations.

2. **Publishable Claim PC2 (strong):** for L-C1 at $d = 6$, $n \geq 4$, the current interior-point solver stack reaches a clear computational ceiling (pass1/pass2 all-timeout pattern), yielding a reproducible deferment criterion for this slice rather than a theorem-level negative statement.

3. **Publishable Claim PC3 (exploratory):** CX-2 exhibits a localized transform-sensitive SONC boundary effect in the Robinson-hat track, confirmed by OP3 targeted sweeps (7×7 local grid and 18-point wide coefficient scan) as a point-level boundary phenomenon at $c = 2.0$, and now motivating a mathematical characterisation of the SONC certificate cone boundary.

Fallback publication path. If OP1 remains unresolved under available solver resources, we package a scope-limited result set centered on PC1 and PC2: robust support-sensitive separation on the frozen slice, explicit solver-ceiling deferment protocol for $d = 6$, $n \geq 4$, and a clearly delimited future-work trajectory for PC3/OP3.

The high-value open problems remain:

1. **Open Problem OP1 (high priority):** determine whether the d=6, $n \geq 4$ L-C1 solver ceiling reflects only algorithmic limits or a deeper structural barrier in SOS decomposition complexity for the tested mean-polynomial families.

2. **Open Problem OP2:** (high priority) formalize support-sensitive SONC membership criteria that explain the persistent $p \in \{1, 2\}$ failures under robust diagnostics while preserving the observed GP feasibility pattern. A prototype classification rule based on support descriptors achieves 48/48 accuracy on the frozen $d \in \{4, 5\}$ nondegenerate slice. It passes all empirical gate criteria (G1–G3). The remaining open step is a theorem-level formulation of the support condition, with either a formal proof or a certified counterexample.

3. **Open Problem OP3 (medium priority):** characterize the localized transform-sensitive SONC boundary behavior in the Robinson-hat family at $c = 2.0$. Targeted OP3 sweeps (a 7×7 local grid and an 18-point wide coefficient scan) establish that the baseline post-transform SONC-feasibility is a *point-level* phenomenon: only $c = 2.0$ succeeds; every tested perturbation fails robustly (H1 Fail, H2 Pass, H3 Pass). The remaining open problem is a mathematical characterisation of the SONC certificate cone boundary at $c = 2.0$, e.g. via Newton-polytope geometry or AM/GM circuit structure analysis.

The current conjecture backlog for integration into the final manuscript is:

- **B1:** state Theorem 3 as a scoped extension claim: computationally confirmed on the nondegenerate $d \in \{4, 5\}$, $n \in \{3, 4\}$ slice and on the partial $d = 6$, $n = 3$ uniform slice; explicitly deferred/open for $d = 6$, $n \geq 4$ under the current SDP solver ceiling, with evidence anchored in the frozen clean summaries, d=6 L-C1 batch summaries, and escalation artifacts.

- **B2:** replace broad SONC containment phrasing by a support-sensitive conjectural statement that is explicitly anchored to the d=4/d=5 robust

$p \in \{1, 2\}$ failure diagnostics, preserves the SONC-feasible subcases on other supports, and keeps the caveat that these computational negatives are slice-local evidence rather than a universal non-membership theorem.

- **B3:** the CX-2 Robinson-hat post-transform SONC-feasibility at $c = 2.0$ is a point-level boundary phenomenon (confirmed by OP3 sweeps). The conjecture is now formalized in the computational synthesis section (B3 paragraph). Escalation from conjectural to theorem-level status requires mathematical characterisation (Newton-polytope or AM/GM analysis), not further solver perturbation.

Next-quarter continuation plan. We define three coordinated tracks for the next-quarter cycle.

1. **Primary path (theory-first):** refine B1 and B2 into publishable theorem/conjecture statements with explicit scope qualifiers, and convert the current computational synthesis into lemma-level support statements that can be cited in the main narrative.

2. **Computational path (solver escalation):** rerun the deferred $d = 6$, $n \geq 4$ L-C1 slices with stronger SDP tooling or structure-exploiting formulations, preserving the same JSONL schema for direct pass-to-pass comparability with the current CVXOPT/SDPA/CSDP baseline.

3. **Fallback path (scope-limited packaging):** if OP1 remains unresolved under available compute, submit a scope-limited manuscript centered on PC1+PC2, with PC3 and OP3 presented as forward-looking conjectural directions and a documented specialized-solver agenda.

Experiment report and reproducibility close-out. For the Phase 4 close-out package, we freeze the experiment-report narrative around three evidence classes: confirmed slices, scope-limited/weakened statements, and anomaly-led forward directions.

- **Confirmed evidence bundle:** frozen d=4/d=5 nondegenerate SOS/GP feasibility with support-sensitive SONC outcomes, plus partial d=6 low-n extension.

- **Deferred evidence bundle:** d=6, $n \geq 4$ L-C1 solver-ceiling deferment documented by pass1/pass2 escalation traces under `CVXOPT->SDPA->CSDP`.

- **Anomaly evidence bundle:** CX-2 localized transform-sensitive boundary behavior and follow-up local-sweep agenda.

Reproducibility is closed out by (i) artifact-path provenance in this section and the theorem ledger, (ii) the generated classification-table snippet inclusion, and (iii) deterministic two-pass manuscript builds via `make pdf` in `MeansResearch/`.

5 Beyond the SOSONC Cone

In this section, we move towards a comprehensive understanding of the general form $M_{q,p}(Y, w)$ beyond the established SOSONC cone. We examine specific classical forms—the Choi-Lam form Q and the Robinson forms R and $\hat{R}$—and establish their representation as mean polynomials.

5.1 Specific Examples

We consider the following forms in $\mathbb{R}[X, Y, Z, W]$:

- The Choi-Lam form Q:

$$Q(X, Y, Z, W) = X^4 + Y^4 + Z^4 + W^4 - 4XYZW$$

- The Robinson form R:

$$\begin{aligned} R(X, Y, Z) = X^6 + Y^6 + Z^6 \\ - (X^4Y^2 + X^4Z^2 + X^2Y^4 + X^2Z^4 + Y^4Z^2 + Y^2Z^4) \\ + 3X^2Y^2Z^2 \end{aligned}$$

- The second Robinson form $\hat{R}$:

$$\begin{aligned} \hat{R}(X, Y, Z, W) = X^2(X - W)^2 + Y^2(Y - W)^2 + Z^2(Z - W)^2 \\ + 2XYZ(X + Y + Z - 2W) \end{aligned}$$

It is observed in [12] that Q is a Sum of Nonnegative Circuit (SONC) polynomial. However, using the Newton polytope method, one can prove that $Q \in \mathcal{P}_{4,4} \setminus \Sigma_{4,4}$ (it is PSD but not SOS).

Remarkably, there exists a linear relationship between Q and $\hat{R}$ as noted by Reznick [11]:

$$\hat{R}(X - W, Y - W, Z - W, X + Y + Z - W) = 2Q(X, Y, Z, W) \tag{1}$$

Since the PSD and SOS cones are closed under linear transformations, the identity above shows that $\hat{R}$ is PSD (respectively SOS) if and only if Q is PSD (respectively SOS). Consequently, $\hat{R}$ also serves as an example of a form in $\mathcal{P}_{4,4} \setminus \Sigma_{4,4}$.

5.2 Mean Polynomial Representations

We now establish the representation of these forms as instances of mean polynomials $M_{q,p}$.

Proposition 3. *The Choi-Lam form Q and the Robinson form $\hat{R}$ can be expressed as mean polynomials of order 1 and 0 (arithmetic minus geometric means).*

Proof. For the Choi-Lam form Q, we observe:

$$Q(X,Y,Z,W) = 4\left(\frac{X^4+Y^4+Z^4+W^4}{4} - \sqrt[4]{X^4Y^4Z^4W^4}\right)$$

This corresponds exactly to the mean form definition with variables being the fourth powers and uniform weights:

$$Q(X,Y,Z,W) = 4M_{1,0}((X^4,Y^4,Z^4,W^4),(1,1,1,1))$$

For the Robinson form $\hat{R}$, applying the inverse of the linear transformation in (1), we derive:

$$\hat{R}(X,Y,Z,W) = 8M_{1,0}((\hat{X}^4,\hat{Y}^4,\hat{Z}^4,\hat{W}^4),(1,1,1,1))$$

where

$$\begin{aligned}(\hat{X},\hat{Y},\hat{Z},\hat{W}) = \frac{1}{2}(&X+W-Y-Z,\ Y+W-X-Z,\\ &Z+W-X-Y,\ W-(X+Y+Z)).\end{aligned}$$

□

This representation is significant for distinguishing the cone of mean polynomials from the SOSONC cone. While Q is SONC (and thus in the SOSONC cone) because it is a mean of monomials in the standard basis, $\hat{R}$ is a mean of *linear forms*. Since the SONC property is not invariant under general linear coordinate changes, $\hat{R}$ serves as a separating form: it belongs to the cone generated by generalized mean polynomials (means of linear forms) but lies outside the standard SOSONC cone (see [12, page 31 and Lemma 4.2.5, page 107]).

6 Positivestellensatz and relation to the cone of $2d$ powers

This section is reserved for a follow-up paper.

7 Application to polynomial optimization

This section is reserved for a follow-up paper.

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Mehdi Ghasemi: mehdi.ghasemi@usask.ca
Edmonton Police Service
Salma Kuhlmann: salma.kuhlmann@uni-konstanz.de
University of Konstanz